

Session C2: Drawing Lewis Structures, Covalent Molecules & Polyatomic Ions

Duration: 90 minutes

Continues from: Session C1, *Electronegativity, Bond Character & the Three Bond Types*. C1 covered **what** a covalent bond is (shared electron pairs, small ΔEN) and how to **name** covalent compounds with prefixes. Today covers **how to draw them**. C1's closing note named Lewis structures as the natural next step.

Grade-9 foundation: NCERT Class 9 Ch9 *Atomic Foundations of Matter*: valence electrons, the octet rule, and the polyatomic-ion list (OH^- , NH_4^+ , NO_3^- , CO_3^{2-} , SO_4^{2-} , HCO_3^-) Dhairya already memorized for the bonding quiz.

Grade-11 depth: NCERT Class 11 Ch4 *Chemical Bonding and Molecular Structure*, §4.1 (Lewis symbols, §4.1.1 octet rule, §4.1.2 covalent bond, §4.1.3 Lewis representation of molecules and ions, §4.1.4 formal charge, §4.1.5 limitations of the octet rule) and §4.3 (bond parameters: bond length, bond enthalpy, §4.3.4 bond order, §4.3.5 resonance). The school's *Gen Chem Lewis* slides (photos) set the exact vocabulary and the 4-step drawing method used below.

Interactive tool: *Lewis Structure Builder*, on the study-tools site under [lewis-builder/](#). It has 20 molecules and ions in 3 levels, and he builds each structure himself with instant hints.

Lesson Plan

Objectives. By the end of this session Dhairya should be able to:

1. Find the number of valence electrons of any main-group atom from its group, and draw its Lewis dot symbol.
2. Tell a **bonding pair** from a **lone pair**, and read single, double and triple bonds as **bond orders** 1, 2 and 3. He should also say which bond is shorter and stronger.
3. Draw the Lewis structure of any simple molecule with the **4-step method**, including molecules that need double or triple bonds.
4. Draw Lewis structures of **polyatomic ions**: adjust the electron count for the charge, draw brackets, and write the charge outside.
5. Use **formal charge** to pick the best structure, and recognize **resonance** (nitrate, carbonate).

Time	Segment
0–8 min	Warm-up: 4 recall questions from C1
8–18 min	Part 1: valence electrons & Lewis dot symbols
18–30 min	Part 2: bonding pairs, lone pairs, bond order, bond length & strength
30–50 min	Part 3: the 4-step method, with worked examples (CH_3OH , CO_2 , HCN , CH_2O)
50–68 min	Part 4: polyatomic ions, formal charge and resonance
68–82 min	Part 5: interactive <i>Lewis Structure Builder</i> (Levels 1 → 3)
82–90 min	Exit ticket, then assign the homework

Pacing note. If you run behind, shorten Part 2's bond-length table (just state "triple is shortest and strongest") and do only Levels 1 and 3 of the builder. Do not cut Part 4. Polyatomic ions are the stated test topic, and the charge adjustment in Step 1 is where most mistakes happen.

Warm-up (8 min)

1. In a covalent bond, what happens to the electrons? → They are **shared** as pairs between two non-metal atoms. The ΔEN is small, below about 1.7.
2. Name CO_2 and N_2O_5 . → Carbon dioxide; dinitrogen pentoxide.
3. Formula and charge of nitrate, carbonate, ammonium and hydroxide? → NO_3^- , CO_3^{2-} , NH_4^+ , OH^- .
4. C1 said the atoms inside a polyatomic ion are held together by what kind of bond? → **Covalent** bonds. The whole group carries one overall charge. So today's method works on ions too.

Bridge to today: "We know CO_2 is *carbon dioxide* and that its atoms share electrons, but *which* electrons, and *how many* bonds? Today we draw it."

Part 1 — Valence Electrons & Lewis Dot Symbols (10 min)

Valence electrons are the electrons in the outermost shell. They are the only ones that take part in bonding.

Shortcut for main-group elements: valence electrons = the group number's last digit (groups 1, 2, 13–18).

Group	1	2	13	14	15	16	17	18
Valence e^-	1	2	3	4	5	6	7	8
Examples	H, Na	Mg, Ca	B, Al	C, Si	N, P	O, S	F, Cl	Ne, Ar
Bonds usually formed	1	2	3	4	3	2	1	0

Lewis dot symbol: the element symbol with its valence electrons as dots. Place them one per side first (4 sides), then pair them up.



Lewis dot symbols

The octet rule (NCERT §4.1.1): atoms bond so that each ends up with **8 electrons** around it, like a noble gas. **Hydrogen** needs only **2** (a *duet*, like helium).

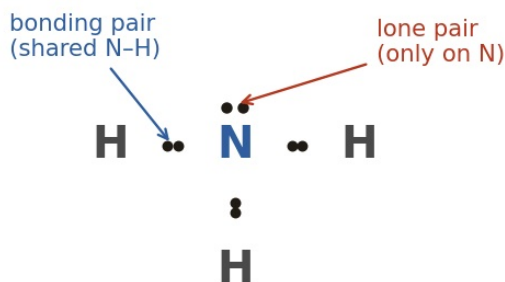
Why "bonds usually formed" works: an atom forms as many bonds as the number of electrons it is *missing* from an octet. N has 5, so it needs 3 more and forms 3 bonds. O has 6, needs 2, and forms 2 bonds. This is the fastest way to check a structure.

Part 2 — Lewis Structures, Bond Pairs & Bond Order (12 min)

Bonding pairs and lone pairs

A **Lewis structure** shows how atoms share some or all of their valence electrons so that each gets an octet (a duet for hydrogen).

- A **bonding pair** is a **shared pair** of electrons, drawn between two atoms. It counts toward the octet (or duet) of **both** atoms.
- A **lone pair** is a pair that belongs to only one atom and is **not involved in bonding**.



Ammonia: three bonding pairs and one lone pair

In NH_3 , nitrogen is surrounded by 3 bonding pairs and 1 lone pair, which is 8 electrons. Each H has 1 bonding pair, which is 2 electrons.

Dashes, and single, double and triple bonds

Each bonding pair is usually drawn as a **dash**, and **each dash = 2 electrons**. Lone pairs stay as dots.

Pairs shared	Bond	Drawn as	Electrons in the bond	Bond order	Example
1	Single	A-B	2	1	F-F, H-O
2	Double	A=B	4	2	O=O, C=O
3	Triple	A≡B	6	3	N≡N, C≡N

Bond order is simply the number of bonds between two atoms.

F-F bond order 1

O=O bond order 2

N≡N bond order 3



Bond order 1, 2, 3

Bond length and bond strength

More shared pairs pull the two nuclei **closer**, and take **more energy** to break. (Numbers from the class slides, for carbon-carbon bonds.)

Bond	Bond order	Bond length	Energy to break
C-C	1	1.535 Å	365 kJ/mol
C=C	2	1.339 Å	598 kJ/mol
C≡C	3	1.203 Å	962 kJ/mol

Rule: as the bond order goes **up**, the bond length goes **down** and the bond strength goes **up**. A triple bond is the shortest and strongest. (1 Å = 10^{-10} m.)

Part 3 — Drawing Lewis Structures: the 4-Step Method (20 min)

This is the method from the class slides, which matches NCERT §4.1.3.

1. **Count the total number of valence electrons.** Add them up for every atom. For a **polyatomic ion**, **add** one electron for each negative charge and **subtract** one for each positive charge.

2. **Connect the atoms with single bonds to make the skeleton.** The **central atom** is the one that forms the most bonds, or the **least electronegative** atom (this is C1's electronegativity trend at work). **Hydrogen and fluorine are never central**, because they can form only one bond. For an ion, draw **square brackets** around the whole structure and write the **charge outside, top right**.
3. **Add lone pairs to the terminal (outer) atoms to complete their octets.** Skip hydrogen, which is already complete with one bond. If any electrons are **left over**, put them as lone pairs on the **central atom**.
4. **If the central atom still lacks an octet, move lone pairs from outer atoms into multiple bonds.** When every atom has its octet (H: 2), the structure is complete.

Always finish with the two checks:

- Electrons drawn = electrons counted in step 1. Each dash counts 2 and each lone pair counts 2.
- Every atom has 8 electrons around it, and every H has 2.

Worked Example 1 — CH₃OH (methanol), the class slide example

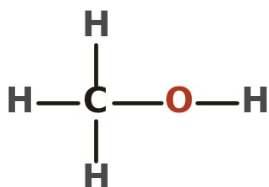
Step 1, count the valence e⁻: C = 4, H = 4 × 1 = 4, O = 6. The total is **14 e⁻**.

Step 2, skeleton: carbon forms 4 bonds, so it is central to its three H and the O. The O bonds to the last H (the formula is written C-H₃ then O-H, which shows the connections). That makes 5 single bonds = 10 e⁻ used, so **4 left**.

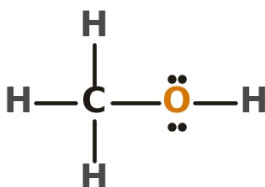
Step 3, lone pairs: the only terminal atom that isn't H is O. It has 2 bonds (4 e⁻), so it needs 4 more: 2 lone pairs. That uses all 4, so **0 left**.

Step 4, check: C has 4 bonds = 8 ✓. O has 2 bonds + 2 lone pairs = 8 ✓. Each H has 1 bond = 2 ✓. Electrons drawn: 5 × 2 + 2 × 2 = 14 ✓.

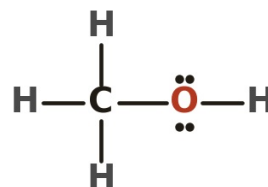
Step 2: skeleton (single bonds)
5 bonds × 2 = 10 e⁻ used, 4 left



Step 3: lone pairs on O
(4 e⁻ → 0 left)



Step 4: check octets
C 8 ✓ O 8 ✓ each H 2 ✓



CH₃OH step by step

Worked Example 2 — CO₂ (needs double bonds)

Step 1: C = 4, O = 2 × 6 = 12, so the total is **16 e⁻**.

Step 2: C is the least electronegative (EN 2.5 vs O 3.5), so it is central: O-C-O. 2 bonds = 4 e⁻, so **12 left**.

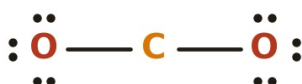
Step 3: each O needs 6 more, which is 3 lone pairs each. That uses all 12, so **0 left**. But **C has only 2 bonds = 4 e⁻. X**

Step 4: move **one lone pair from each O** into the bond with C. Each single bond becomes a double bond: **O=C=O**. Now C has 2 double bonds = 8 ✓, and each O has 1 double bond + 2 lone pairs = 8 ✓. The total is still 16 ✓.

Step 2: skeleton
2 bonds = 4 e⁻ used, 12 left



Step 3: 3 lone pairs on each O
0 left, but C has only 4 e⁻ X



Step 4: move 1 pair from each O
into a bond: C 8 ✓, O 8 ✓



CO₂ step by step

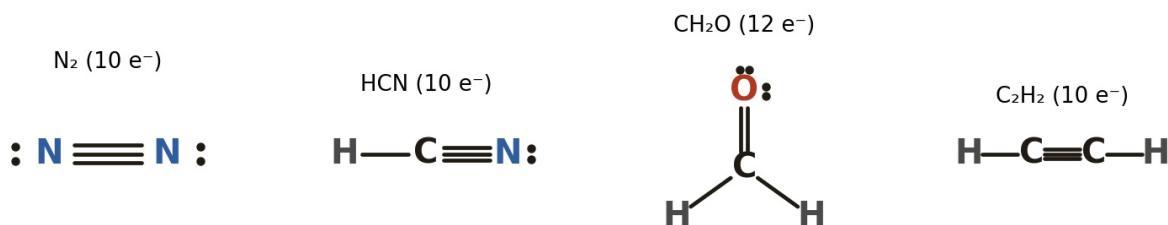
Key idea: step 4 never adds electrons. It only **moves** an existing lone pair so that it becomes shared. That is why a double bond appears exactly when the electrons “run out” before the central atom is full.

Worked Example 3 — HCN (needs a triple bond)

- **Step 1:** H = 1, C = 4, N = 5, so the total is **10 e⁻**.
- **Step 2:** H can't be central. C forms the most bonds, so the skeleton is H-C-N. 2 bonds = 4 e⁻, so **6 left**.
- **Step 3:** N gets 3 lone pairs (6 e⁻), so **0 left**. C has only 2 bonds = 4 e⁻. **X**
- **Step 4:** move **two** lone pairs from N into the C-N bond, which becomes **C≡N**. Now C has 1 + 3 = 4 bonds = 8 ✓ and N has 3 bonds + 1 lone pair = 8 ✓.
- **Result:** H-C≡N: with one lone pair on N.

Worked Example 4 — CH₂O (formaldehyde)

- **Step 1:** 4 + 2(1) + 6 = **12 e⁻**.
- **Step 2:** C is central, bonded to O and both H. 3 bonds = 6 e⁻, so **6 left**.
- **Step 3:** O gets 3 lone pairs, so **0 left**. C has 3 bonds = 6 e⁻. **X**
- **Step 4:** move one O lone pair into the C-O bond, making **C=O**. Now C has 8 ✓ and O has 2 bonds + 2 lone pairs = 8 ✓.



More molecules

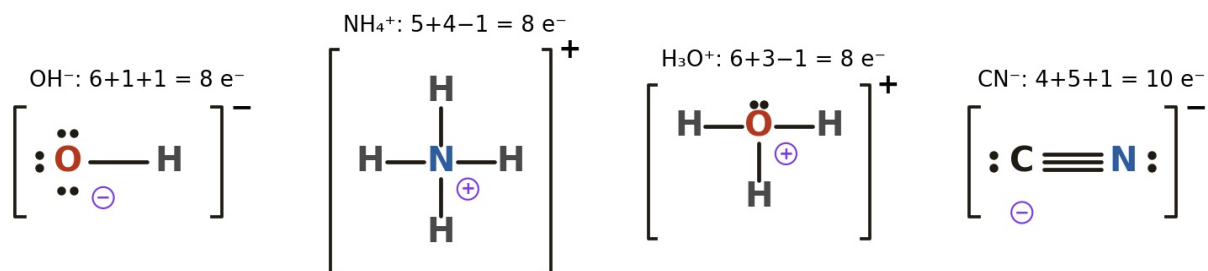
Pattern to notice: C always ends with **4 bonds**, N with **3 bonds + 1 lone pair**, O with **2 bonds + 2 lone pairs**, and a halogen with **1 bond + 3 lone pairs**. This matches the “bonds usually formed” row in Part 1 and is a fast way to check any neutral molecule.

Part 4 — Polyatomic Ions, Formal Charge & Resonance (18 min)

What changes for an ion

A polyatomic ion is a group of covalently bonded atoms with an **overall charge** (C1). The charge means electrons were **gained** or **lost**, so:

- **Step 1 changes:** a **negative** charge adds electrons and a **positive** charge removes them. OH⁻ has 1 extra electron; NH₄⁺ has 1 fewer.
- **The drawing changes:** put **square brackets** around the whole structure and write the charge **outside, top right**.



Simple polyatomic ions

Ion	Step 1 count	Structure
OH ⁻	6 + 1 + 1 = 8	O-H with 3 lone pairs on O
NH ₄ ⁺	5 + 4 - 1 = 8	N with 4 single bonds to H, no lone pairs
H ₃ O ⁺	6 + 3 - 1 = 8	O with 3 bonds to H + 1 lone pair
CN ⁻	4 + 5 + 1 = 10	C≡N with 1 lone pair on each atom

Formal charge: where the charge “sits”

The purple circled signs in the figures are **formal charges** (NCERT §4.1.4). A formal charge compares how many electrons an atom “owns” in the structure with how many valence electrons it brought:

$$\text{Formal charge} = V - L - \frac{1}{2}B$$

where V = valence electrons of the free atom, L = electrons in its **lone pairs**, and B = electrons in its **bonds** (so $\frac{1}{2}B$ = the number of bonds).

Why it works: the atom keeps all of its lone-pair electrons and **half** of each shared pair. If it ends up owning fewer electrons than it brought, it is positive; if more, negative.

Checks:

- The formal charges always **add up to the overall charge** of the molecule or ion.
- The **best** structure has formal charges as close to **zero** as possible. Any negative charge should be on the **more electronegative** atom.

Worked check — OH⁻: O has V = 6, L = 6 (3 lone pairs) and 1 bond, so FC = 6 - 6 - 1 = **-1**. H has 1 - 0 - 1 = 0. The sum is -1, which matches the charge ✓. The negative charge sits on the O.

Worked check — NH₄⁺: N has 5 - 0 - 4 = **+1**, and each H is 0. The sum is +1 ✓.

Using formal charge to choose between structures (CO₂): both of these give every atom an octet, but only one is right:

O=C=O: formal charges 0, 0, 0 ✓ best O≡C-O: octets OK, but +1 and -1 ✗



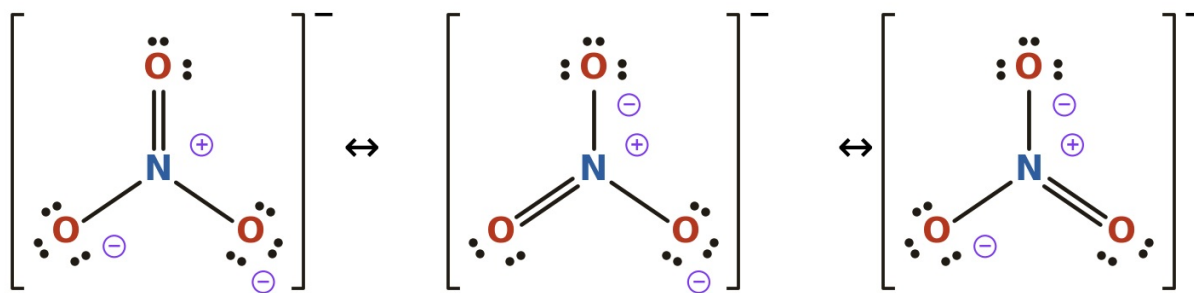
Formal charge picks the best structure

O=C=O has all formal charges 0, so it is the correct structure. O≡C-O puts +1 and -1 on the two oxygens, so it is worse.

Worked Example 5 — NO₃⁻ (nitrate) and resonance

- Step 1:** N = 5, O = 3 × 6 = 18, plus **1** for the charge = **24 e⁻**.
- Step 2:** N is less electronegative than O (3.0 vs 3.5), so N is central with 3 O around it. 3 bonds = 6 e⁻, so **18 left**. Draw brackets with a - charge.
- Step 3:** each O gets 3 lone pairs (18 e⁻), so **0 left**. N has 3 bonds = 6 e⁻. ✗
- Step 4:** move one lone pair from **one** O into a double bond with N. Now N has 8 ✓, the double-bonded O has 2 lone pairs, and the other two O each have 3.
- Formal charges:** N = 5 - 0 - 4 = **+1**. The O in N=O is 6 - 4 - 2 = 0. Each O in N-O is 6 - 6 - 1 = **-1**. The sum is +1 - 1 - 1 = **-1** ✓.

But which O gets the double bond? Any of the three: the structures are equally good. When more than one valid Lewis structure differs **only in where the double bond (electrons) is placed**, they are called **resonance structures** and are joined with a double-headed arrow ↔. (Compare the ozone and nitrate diagrams in the class slides.)

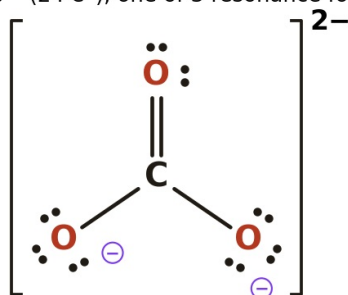


Nitrate resonance structures

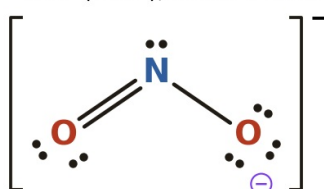
What it really means: the real ion is a **blend** of all three. Experiments show all three N-O bonds are **identical**, each between a single and a double bond (bond order $1\frac{1}{3}$). The double bond isn't jumping around; no single drawing can show it alone.

Worked Example 6 — CO_3^{2-} , NO_2^- and SO_4^{2-}

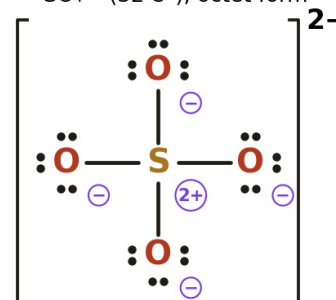
CO_3^{2-} (24 e^-), one of 3 resonance forms



NO_2^- (18 e^-), one of 2 forms



SO_4^{2-} (32 e^-), octet form



Carbonate, nitrite, sulfate

- CO_3^{2-} : $4 + 18 + 2 = 24 e^-$. The same shape and steps as nitrate: one $\text{C}=\text{O}$ and two $\text{C}-\text{O}$. Each single-bonded O has FC -1 , so the sum is $-2 \checkmark$. It has 3 resonance forms.
- NO_2^- : $5 + 12 + 1 = 18 e^-$. The skeleton $\text{O}-\text{N}-\text{O}$ uses 4, leaving 14. Each O takes 3 lone pairs (12), leaving 2, which go on **N** as a lone pair (step 3's "leftover electrons go on the central atom"). N then has 2 bonds + 1 lone pair = 6 $\times$. Move one O lone pair to make $\text{N}=\text{O}$, and N has 8 $\checkmark$. There are 2 resonance forms.
- SO_4^{2-} : $6 + 24 + 2 = 32 e^-$. 4 single bonds ($8 e^-$) + 3 lone pairs on each O ($24 e^-$) = 32, so every atom already has an octet and **no step 4 is needed**. The formal charges are $\text{S} = 6 - 0 - 4 = +2$ and each $\text{O} = -1$, which sum to $-2 \checkmark$.
 - *Beyond the octet (for later)*: sulfur is in period 3 and can hold more than 8 electrons (an **expanded octet**, NCERT §4.1.5). Some textbooks draw SO_4^{2-} with two $\text{S}=\text{O}$ bonds to reduce the formal charges. At this level, the octet form above is the expected answer. Follow your teacher's version.

When the octet rule bends (NCERT §4.1.5, awareness only)

Exception	Example	What happens
Incomplete octet	BF_3 , BeCl_2	B has only 6 e^- , Be only 4
Odd number of electrons	NO , NO_2	Someone has to be left with 7
Expanded octet (period 3 and below)	PCl_5 , SF_6	P has 10 e^- , S has 12

Part 5 — Interactive Activity: Lewis Structure Builder (14 min)

Open **study tools** → **Chemistry** → **Lewis Structure Builder**. It follows exactly the 4 steps above:

1. **Count:** type the total valence electrons. A wrong answer shows the per-atom breakdown, including

the charge adjustment for ions.

2. **Skeleton:** pick the central atom. A wrong pick explains the rule (least electronegative, never H or F).
3. **Build:** tap bonds to switch between single, double and triple, and tap atoms to add or remove lone pairs. A live counter shows electrons **left to place**, and every atom shows its own electron count (e.g. "6/8").
4. **Check:** the builder says exactly what is wrong (e.g. "N has only 6 electrons"). If the octets are right but the formal charges could be smaller, it says so. For nitrate or carbonate, **any** resonance form is accepted.

Suggested run for today: Level 1 → H₂O, NH₃, CH₃OH; Level 2 → CO₂, HCN, CH₂O; Level 3 → OH⁻, NH₄⁺, NO₃⁻, SO₄²⁻. Let him use hints freely on the first of each level and none on the last. The stars show how independent he was.

Exit Ticket (8 min)

1. How many valence electrons does SO₃²⁻ have in total? → 6 + 3(6) + 2 = **26**.
2. Why can't hydrogen be the central atom? → It can form only **one** bond (it needs just 2 electrons).
3. Which is shorter and stronger, C=O or C≡O? → **C≡O**. A higher bond order gives a shorter, stronger bond.
4. In NO₃⁻, why are there three drawings joined by ↔? → **Resonance**. The double bond could be drawn to any of the 3 O; the real ion is an average, with all three bonds equal.
5. What is the formal charge on N in NH₄⁺? → 5 - 0 - 4 = **+1**.

Homework — Draw These (answers below)

For each one: show the 4 steps, count the electrons, and write the formal charge on every atom that isn't zero. Use brackets and a charge for ions.

1. H₂S
2. PCl₃
3. C₂H₄ (the two C atoms are bonded to each other)
4. NO⁺
5. ClO⁻ (hypochlorite)
6. SO₃²⁻ (sulfite)
7. PO₄³⁻ (phosphate), octet form

Answer Key — Full Step-by-Step Solutions

1. H₂S

- Step 1: S = 6, H = 2 × 1, so the total is **8 e⁻**.
- Step 2: S is central (H never is): H-S-H. 2 bonds = 4 e⁻, so 4 left.
- Step 3: the terminal atoms are both H, so they are skipped. The 4 leftover electrons go on S as 2 lone pairs.
- Step 4: S has 2 bonds + 2 lone pairs = 8 ✓, and each H has 2 ✓. All formal charges are 0.

Answer: H-S-H with 2 lone pairs on S (the same pattern as water: S is in the same group as O).

2. PCl₃

- Step 1: P = 5, Cl = 3 × 7 = 21, so the total is **26 e⁻**.
- Step 2: P is central (less electronegative than Cl). 3 bonds = 6 e⁻, so 20 left.
- Step 3: each Cl gets 3 lone pairs (18 e⁻), so 2 left. Those go on P as 1 lone pair.

- Step 4: P has 3 bonds + 1 lone pair = 8 ✓, and each Cl has 8 ✓. The formal charges are $P = 5 - 2 - 3 = 0$ and $Cl = 7 - 6 - 1 = 0$.

Answer: P with 3 single bonds to Cl, 1 lone pair on P and 3 on each Cl.

3. C₂H₄

- Step 1: $2 \times 4 + 4 \times 1 = 12 \text{ e}^-$.
- Step 2: C-C with 2 H on each C. 5 bonds = 10 e⁻, so 2 left.
- Step 3: the only terminal atoms are H. The 2 leftover electrons go on one C as a lone pair, which gives that C 8 but leaves the other C with 3 bonds = 6 ✗.
- Step 4: move that lone pair into the C-C bond to make **C=C**. Now each C has 1 double + 2 single = 4 bonds = 8 ✓.

Answer: H₂C=CH₂, with no lone pairs.

4. NO⁺

- Step 1: $5 + 6 - 1 = 10 \text{ e}^-$.
- Step 2: N-O. 1 bond = 2 e⁻, so 8 left.
- Step 3: 3 lone pairs on O (6 e⁻), so 2 left. Those go on N as 1 lone pair, but N then has only 4 ✗.
- Step 4: move 2 lone pairs from O into the bond, giving **N≡O**. Each atom now has 3 bonds + 1 lone pair = 8 ✓.
- Formal charges: $N = 5 - 2 - 3 = 0$ and $O = 6 - 2 - 3 = +1$. The sum is +1 ✓.

Answer: [:N≡O:]⁺.

5. ClO⁻

- Step 1: $7 + 6 + 1 = 14 \text{ e}^-$.
- Step 2: Cl-O. 2 e⁻ used, so 12 left.
- Step 3: 3 lone pairs on O (6), then 3 lone pairs on Cl (6), so 0 left.
- Step 4: Cl has 8 ✓ and O has 8 ✓.
- Formal charges: $Cl = 7 - 6 - 1 = 0$ and $O = 6 - 6 - 1 = -1$. The sum is -1 ✓.

Answer: [Cl-O]⁻ with 3 lone pairs on each atom.

6. SO₃²⁻

- Step 1: $6 + 18 + 2 = 26 \text{ e}^-$.
- Step 2: S is central with 3 O. 3 bonds = 6 e⁻, so 20 left.
- Step 3: each O gets 3 lone pairs (18), so 2 left. Those go on S as 1 lone pair.
- Step 4: S has 3 bonds + 1 lone pair = 8 ✓, and each O has 8 ✓.
- Formal charges: $S = 6 - 2 - 3 = +1$ and each $O = 6 - 6 - 1 = -1$. The sum is $+1 - 3 = -2$ ✓.

Answer: as in the figure below.

7. PO₄³⁻

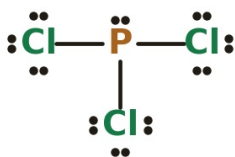
- Step 1: $5 + 24 + 3 = 32 \text{ e}^-$.
- Step 2: P is central with 4 O. 4 bonds = 8, so 24 left.
- Step 3: 3 lone pairs on each O uses all 24, so 0 left.
- Step 4: P has 4 bonds = 8 ✓ and each O has 8 ✓.
- Formal charges: $P = 5 - 0 - 4 = +1$ and each $O = -1$. The sum is $+1 - 4 = -3$ ✓.

Answer: the same shape as sulfate.

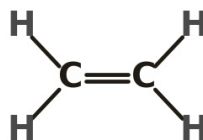
1. H_2S (8 e^-)



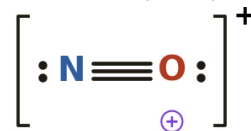
2. PCl_3 (26 e^-)



3. C_2H_4 (12 e^-)

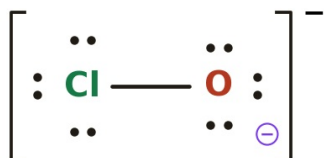


4. NO^+ (10 e^-)

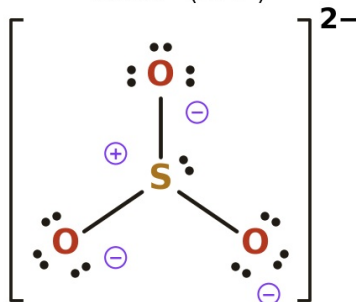


Homework answers 1-4

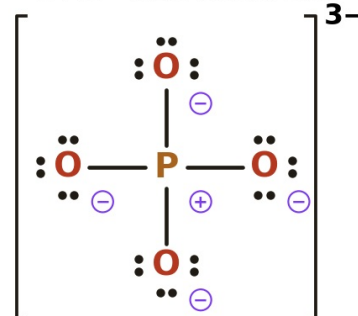
5. ClO^- (14 e^-)



6. SO_3^{2-} (26 e^-)



7. PO_4^{3-} (32 e^-), octet form



Homework answers 5-7

Notes for next session

Lewis structures show **which** electrons are shared, but not the **3-D shape**. The next step is **VSEPR theory** (NCERT Class 11 Ch4 §4.4). The lone pairs drawn today push bonding pairs apart. That is why H_2O is **bent** (104.5°) while CO_2 is **linear**, and why NH_3 is a pyramid. It also finally explains **molecular polarity**: CO_2 has polar $\text{C}=\text{O}$ bonds but is a non-polar molecule, because the two dipoles cancel.